

IMO Training – Number Theory

Level 2 Assignment 1 Solutions

1. If $m = n$, then $\text{LHS} = 0$, so $(m, n) = (0, 0)$.
 Otherwise, let $d = (m, n)$, then $m = dM, n = dN, (M, N) = 1$.
 Then the equation becomes $d^3(M^5 - N^5) = 16MN$.
 Note that $(M^5 - N^5, MN) = 1$, so $M^5 - N^5 \mid 16$.
 If one of $|M|, |N| \geq 2$, then the difference of fifth powers will exceed 16. So $|M|, |N| \leq 1$.
 Solving, we have $(M, N) = (-1, 1), d = 2$. So $(m, n) = (-2, 2)$.
 Checking: $0^5 - 0^5 = 16 \cdot 0 \cdot 0 = 0; (-2)^5 - 2^5 = 16(-2)(2) = -64$.

2. Take mod 5 on both sides, we have $3^m \equiv 4 \pmod{5}$. This implies $m \equiv 2 \pmod{4}$.
 On the other hand, if $n \geq 4$, then take mod 16 on both sides, $3^m \equiv 1 \pmod{16}$.
 This implies $m \equiv 0 \pmod{4}$. Contradiction.
 So n is at most 3. Looking at remaining cases we have $(m, n) = (6, 3)$.
 Checking: $125 \cdot 2^3 - 3^6 = 1000 - 729 = 271$.

3. Introduce a lemma:
Lemma. For $a \leq b$, $2^{2^{a+1}} + 2^{2^a} + 1$ divides $2^{2^{b+1}} + 2^{2^b} + 1$.
Proof. Note that $2^{2^{a+2}} + 2^{2^{a+1}} + 1 = (2^{2^{a+1}} + 2^{2^a} + 1)(2^{2^{a+1}} - 2^{2^a} + 1)$.
 And by writing $b = a + k, k \geq 0$, the lemma is proved by Induction on k . □

 Claim that all $n = 2^{2^k}$, where $k \in \mathbb{N}$, satisfy the requirement.
 If this claim holds, then the statement holds as there are infinitely such n .
 Indeed, for such n , $4^n + 2^n + 1 = 2^{2^{2^k+1}} + 2^{2^{2^k}} + 1, n^2 + n + 1 = 2^{2^{k+1}} + 2^{2^k} + 1$.
 Since $k < 2^k$ for all $k \geq 1$, the claim holds by the Lemma. We are done :)

4. For $2^x + 3^y = n^2$, some $n \in \mathbb{N}$, take mod 3 on both sides, $(-1)^x \equiv n^2 \pmod{3}$.
 So $(-1)^x \equiv 1 \pmod{3}$, x even. So write $x = 2k$, some $k \in \mathbb{N}$.
 Now $3^y = (n - 2^k)(n + 2^k)$. Both $n - 2^k$ and $n + 2^k$ are powers of 3,
 with difference being a power of 2, in particular not divisible by 3.
 So the only possibility is $n - 2^k = 1, 3^y - 1 = 2^{k+1}$. Rewriting as $3^y - 2^{k+1} = 1$, we can
 deduce $y = k = 2$ by Catalan.
 So $(x, y) = (4, 2)$. Checking: $2^4 + 3^2 = 25 = 5^2$.

5. If $t = 0$, then the equation becomes $1 + 3^x 4^y = z^2$.

If $y = 0$, then $1 + 3^x = z^2$, which gives $(z + 1)(z - 1) = 3^x$.

Since difference of two powers of 3 is 2, one is 3 and the other is 1.

So $(t, x, y, z) = (0, 1, 0, 2)$. Checking, $5^0 + 3^1 \cdot 4^0 = 4 = 2^2$.

If $y > 0$, then $3^x 2^{2y} = (z + 1)(z - 1)$. Both $z + 1$ and $z - 1$ are product of powers of 2 and 3 only, with difference exactly 2. So one of the two terms must be a power of 2.

Now for $a + b = 2y$, divide into two cases:

$$\begin{cases} z + 1 = 3^x \cdot 2^a \\ z - 1 = 2^b \end{cases} \iff 2^b + 2 = 3^x \cdot 2^a.$$

So $a = 1$. The equation becomes $2^{b-1} + 1 = 3^x$, and hence $(x, b) = (2, 4), (1, 2)$

The two solutions are rejected as $a + b$ is odd. On the other hand,

$$\begin{cases} z - 1 = 3^x \cdot 2^a \\ z + 1 = 2^b \end{cases} \iff 2^b - 2 = 3^x \cdot 2^a.$$

Again $a = 1$. The equation becomes $2^{b-1} - 1 = 3^x$, and hence $x = 1, b = 3$.

So $(t, x, y, z) = (0, 1, 2, 7)$. Checking, $5^0 + 3^1 \cdot 4^2 = 49 = 7^2$.

If $t > 0$ and $x = 0$, then the equation becomes $5^t + 4^y = z^2$.

Consequently, $5^t = (z - 2^y)(z + 2^y)$. Since both terms in RHS are powers of 5, with difference not divisible by 5, we have $z - 2^y = 1$. So $2^{y+1} = 5^t - 1$.

Since 3 divides $5^t - 1$ when t even and 8 does not divide $5^t - 1$ when $t > 1$ odd, so $t = 1, y = 1$.

Solving, $(t, x, y, z) = (1, 0, 1, 3)$. Checking, $5^1 + 3^0 \cdot 4^1 = 9 = 3^2$.

If $t, x > 0$ and $y = 0$, then the equation becomes $5^t + 3^x = z^2$.

Take mod 4 on both sides, we have $1 + (-1)^x \equiv z^2 \pmod{4}$, which we must have x odd.

Take mod 10 on both sides, as $3^x \equiv 3, 7 \pmod{10}$ for x odd, $z^2 \equiv 2, 8 \pmod{10}$.

No solution.

If $t, x, y > 0$,

Take mod 3 on both sides, $(-1)^t \equiv z^2 \pmod{3}$, and so t must be even. Let $t = 2u, u \in \mathbb{N}$.

Take mod 5 on both sides, $3^x (-1)^y \equiv z^2 \pmod{5}$, and so $3^x \equiv \pm 1 \pmod{5}$, i.e. x even.

Let $x = 2v, v \in \mathbb{N}$. Then the equation becomes $5^{2u} + 3^{2v} 2^{2y} = z^2$.

Rewrite and we have $5^{2u} = (z - 3^v 2^y)(z + 3^v 2^y)$. By similar argument, $z - 3^v 2^y = 1$.

So $1 + 3^v 2^{y+1} = 5^{2u}$, $3^v 2^{y+1} = (5^u - 1)(5^u + 1)$. Hence $\{5^u - 1, 5^u + 1\} = \{2^y, 2 \cdot 3^v\}$.

Divide into cases:

$$\begin{cases} 5^u + 1 = 2^y \\ 5^u - 1 = 2 \cdot 3^v \end{cases} \iff 2^{y-1} - 3^v = 1.$$

Solving, $y = 3, v = 1$. But $5^u = 7$, reject. On the other hand,

$$\begin{cases} 5^u - 1 = 2^y \\ 5^u + 1 = 2 \cdot 3^v \end{cases} \iff 3^v - 2^{y-1} = 1.$$

We have $(v, y) = (1, 2), (2, 4)$. For $(v, y) = (2, 4)$, $5^u = 17$, reject.

For $(v, y) = (1, 2)$, $u = 1$. $\therefore (t, x, y, z) = (2, 2, 2, 13)$. Checking, $5^2 + 3^2 \cdot 4^2 = 169 = 13^2$.

To conclude, $(t, x, y, z) = (0, 1, 0, 2), (0, 1, 2, 7), (1, 0, 1, 3), (2, 2, 2, 13)$.

6. For such n , if $n \geq 2$, and if the corresponding prime p is not 3, then divide into two cases:
 Case 1: If $p \equiv 1 \pmod{3}$, then $p^n - (p-1)^n \equiv 1 \pmod{3}$, and since $n \geq 2$, it surely exceeds 1, hence cannot be a power of 3.
 Case 2: If $p \equiv 2 \pmod{3}$, then $p^n - (p-1)^n \equiv 2^n - 1 \pmod{3} \equiv 0 \pmod{3}$, so n even.
 Write $n = 2k, k \in \mathbb{N}$. Letting $p^n - (p-1)^n = 3^m$, some $m \in \mathbb{N}$, we can rewrite it as

$$(p^k + (p-1)^k)(p^k - (p-1)^k) = 3^m.$$

Now let $\begin{cases} p^k + (p-1)^k = 3^a \\ p^k - (p-1)^k = 3^b \end{cases}$, we have $2 \cdot p^k = 3^a + 3^b$. Consider mod 3, we have $b = 0$.

Now $p^k - (p-1)^k = 1$. If $k \geq 2$, then $p^k - (p-1)^k > p - (p-1) = 1$. So $k = 1, n = 2$.

If $n = 1$, any prime works as the difference is simply $1 = 3^0$.

Checking for $n = 2$: $2^2 - 1^2 = 3^1$.

7. The answer is 504.

First we show that the answer cannot exceed 504. Directly compute $(11^6 - 1, 13^6 - 1) = 504$.
 Next, we show that 504 indeed divides $p^6 - 1$ for all primes $p > 7$.

Indeed, notice that $504 = 7 \times 8 \times 9$.

By Fermat-little Theorem, $p^6 \equiv 1 \pmod{7}$, so 7 divides $p^6 - 1$.

By Fermat-Euler Theorem, since 9 is coprime with all primes $p > 7$, $p^6 \equiv 1 \pmod{9}$,
 so 9 divides $p^6 - 1$.

Finally, for all primes $p > 7$, p odd. By considering $p \equiv 1, 3, 5, 7 \pmod{8}$, $p^6 \equiv 1 \pmod{8}$.
 So the claim holds. Done.

8. Consider equation $\frac{(a-b)(b-c)(c-a)}{2} + 2 = 2016^k$, where $a, b, c, k \in \mathbb{Z}, k \geq 0$.

Let $x = a - b, y = b - c$, then $c - a = -(x + y)$.

Simplify and the equation becomes $-xy(x + y) = 2 \cdot 2016^k - 4$.

If $k > 0$, take mod 7 on both sides, $xy(x + y) \equiv 4 \pmod{7}$.

Note that the expression equals $x^2y + y^2x$, and the product of the two terms is a cube.

For sum of two terms congruent to 4 (mod 7), the two terms must be either $\{1, 3\}, \{2, 2\}$,
 or $\{5, 6\} \pmod{7}$. In such cases, product has to be congruent to 2, 3, 4 (mod 7).

However, cubes must be congruent to $\pm 1, 0 \pmod{7}$. So no solution in this case.

So it remains to consider $xy(x + y) = 2$. By checking one by one, the only integral solution
 is $(x, y) = (1, 1)$. Hence by symmetry, any a, b, c s.t. $\{a - b, b - c, c - a\} = 1, 1, -2$ works.

$(a, b, c) = (t + 2, t + 1, t)$ and its cyclic permutations, where $t \in \mathbb{Z}$.

9. The answer is 10.

By pigeonhole, we will have at least two numbers with same remainder when divided by 11
 if there are more than 11 numbers. So the required answer cannot exceed 11. Furthermore,
 if there are 11 numbers, they must form a complete residue system mod 11. WOLOG let
 $11 \mid x_1$. Then the expression in (b) $\equiv x_2 \cdots x_{10} \pmod{11} \equiv -1 \pmod{11}$, by Wilson.

So the required answer cannot exceed 10. Now consider $\{x_1, \dots, x_{10}\} = \{1, \dots, 10\}$.

It is obvious that no pairwise difference is divisible by 11, and the required sum is congruent
 to sum of inverses of 1 to 10 mod 11. We know that as 11 is prime, inverses of 1 to 10 mod
 11 is simply a shuffle of the complete residue system mod 11.

So the sum must be divisible by 11. Done.

Remark. These 9 problems are extracted from shortlisted problems for Junior Balkan Mathematical Olympiad (JBMO). This contest is open for school students of age under 15.5 (F4/G10 or below for Hong Kong) from Balkan countries. Contestants are given 4.5 hours to solve 4 problems. This is a very good starting point to practice problems in international-level MO contests.